

CLAYTON LAKE, ME, USA

WMO: 727119

Lat: 46.617N

Lon: 69.533W

Elev: 1031

StdP: 14.16

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94605

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-15.3	-32.5	0.9	-5.8	-26.0	1.3	-5.9	16.2	33.8	13.5	27.4	1.2	280	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	83.3	66.7	80.6	65.1	77.9	63.9	70.1	78.1	68.5	75.9	67.0	74.4	6.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	106.0	72.6	66.0	99.4	72.3	64.0	92.6	71.4	34.7	78.6	33.4	75.5	32.1	74.3	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.1	10.6	-28.0	88.3	6.0	1.6	-32.2	89.5	-35.7	90.4	-39.1	91.3	-43.4	92.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	8.3	12.5	23.3	36.0	49.0	58.6	64.4	63.2	54.6	43.5	30.7	16.4		
	DBStd	21.41	13.55	12.42	11.39	8.15	7.40	6.88	5.45	5.58	7.95	7.96	9.53	11.74		
	HDD50	5607	1293	1049	828	424	109	7	0	0	40	235	579	1043		
	HDD65	9812	1758	1469	1292	870	499	214	79	102	325	667	1029	1508		
	CDD50	1415	0	0	1	3	78	266	447	409	177	33	1	0		
	CDD65	144	0	0	0	0	3	23	61	45	12	0	0	0		
	CDH74	1599	0	0	0	2	63	322	620	460	128	4	0	0		
	CDH80	277	0	0	0	0	10	66	107	76	18	0	0	0		
	WSAvg	3.8	3.9	4.1	4.1	4.3	4.1	3.6	3.5	3.2	3.2	3.7	3.9	3.9		
	PrecAvg	39.6	2.7	2.3	2.6	3.0	3.1	3.9	4.6	3.6	3.7	3.9	3.1	3.4		
(15) Precipitation	PrecMax	46.9	4.4	3.9	5.0	6.3	5.5	5.9	6.7	7.2	8.4	8.5	5.3	7.6		
	PrecMin	31.2	0.5	1.1	1.1	1.2	1.1	1.3	2.4	0.2	2.0	1.0	1.7	1.6		
	PrecStd	4.6	1.2	0.7	0.9	1.5	1.1	1.3	1.3	1.7	1.7	1.9	1.0	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	48.2	57.2	68.8	81.0	86.2	86.3	85.5	82.3	72.5	58.3	46.3		
		MCWB	42.3	43.5	46.5	52.4	62.9	67.6	68.3	66.3	67.0	55.8	49.8	43.2		
	2%	DB	38.9	40.9	47.8	61.0	74.5	81.5	83.1	82.1	77.7	66.1	52.9	40.0		
		MCWB	36.8	36.5	38.2	48.7	58.3	64.9	66.9	65.8	64.6	53.1	46.1	37.7		
	5%	DB	33.8	36.1	43.2	55.2	69.1	77.8	80.2	79.1	73.4	61.6	48.4	36.4		
		MCWB	31.5	32.8	35.3	44.8	55.4	62.6	65.6	64.3	62.5	51.5	43.6	34.1		
	10%	DB	29.3	33.0	39.3	50.1	64.5	73.5	77.5	76.2	69.6	57.3	44.5	33.6		
		MCWB	27.4	30.6	33.8	41.4	53.0	60.2	64.1	63.9	59.4	48.7	41.2	31.7		
	0.4%	WB	41.5	44.2	50.2	57.4	66.7	70.5	73.1	71.3	70.4	63.4	53.4	43.4		
		MCDB	43.1	47.6	55.3	66.4	74.9	81.4	80.5	80.3	77.7	68.2	56.7	45.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.0	37.0	40.1	50.3	62.0	68.3	70.5	69.3	67.8	58.1	47.6	37.0		
		MCDB	35.9	40.0	44.6	58.6	70.6	77.7	78.0	76.6	73.3	64.0	51.2	38.5		
	5%	WB	30.1	33.3	36.7	46.1	58.4	65.7	68.6	67.8	65.3	53.9	44.3	33.7		
		MCDB	32.1	35.5	42.0	53.2	65.9	74.4	76.0	74.8	71.9	59.2	47.4	35.4		
	10%	WB	25.9	30.4	33.9	42.4	55.0	63.1	67.0	66.1	62.4	50.4	41.0	31.1		
		MCDB	27.6	33.1	38.7	49.7	63.0	71.4	74.7	73.9	69.0	55.8	44.4	32.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	23.6	23.2	20.2	22.8	23.1	22.1	22.7	23.0	17.5	14.8	18.2		
		MCDBR	20.7	23.3	28.0	30.4	31.2	29.1	25.7	26.6	26.6	25.8	20.2	16.1		
		MCWBR	21.6	21.2	20.0	19.1	17.2	15.3	12.5	13.6	14.5	17.5	15.4	14.2		
	5% WB	MCDBR	20.9	22.2	22.1	27.9	26.3	23.6	21.6	20.9	20.7	22.8	17.5	14.8		
		MCWBR	20.2	20.1	16.5	19.2	16.0	13.4	11.4	10.8	12.0	17.0	15.2	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.265	0.315	0.332	0.352	0.384	0.398	0.381	0.358	0.327	0.292	0.266		
	taud		2.386	2.312	2.254	2.411	2.426	2.328	2.306	2.363	2.418	2.505	2.527	2.408		
	Ebn at Noon		276	291	288	293	290	279	273	274	271	265	257	253		
	Edh at Noon		22	29	36	35	35	39	39	36	31	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		446	727	1107	1427	1593	1730	1729	1527	1212	721	414	313		
	RadStd		58	85	75	122	171	111	154	93	88	65	46	27		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page